BlockBase - Day 4

Mentored by FEC

What is Gas?

What is Ethereum Gas? (Examples + Easy Explanation)

Ethereum is the second largest crypto by market cap after Bitcoin and the leading blockchain platform known for its decentralized applications (dApps) and smart contracts. A crucial component of Ethereum's functionality is gas fees. Gas fees are payments made by users to compensate for the computing energy required to process and validate transactions on the Ethereum network. Understanding gas fees is essential for anyone using Ethereum, as they directly impact the cost and efficiency of transactions.

Gas fees on Ethereum represent the cost of performing transactions or executing smart contracts on the network. These fees are paid in Ether (ETH), Ethereum's native cryptocurrency. Gas is a unit that measures the amount of computational effort required to execute operations. The more complex the operation, the higher the gas required.

Gas fees are calculated using two main components: gas units and gas price. Gas units measure the amount of work needed for a transaction, while the gas price (denoted in gwei) determines how much you pay per unit of gas. One gwei is equal to 0.000000001 ETH.

Consider you want to transfer ETH to another wallet. A simple ETH transfer typically requires 21,000 gas units. If the gas price is set at 20 gwei due to current network conditions, the total gas fee would be 21,000 * 20 gwei = 420,000 gwei, or 0.00042 ETH. If network congestion increases, the gas price might rise, making the transaction more expensive.

▶️ Introduction to gas
▶️ Gas in depth

Factors Influencing Ethereum (ETH) Gas Fees

- **Network Demand:** Network demand significantly influences gas prices. When many users are trying to process transactions simultaneously, the gas price increases. This is because users compete to have their transactions included in the next block, incentivizing miners by offering higher gas prices. Conversely, when network activity is low, gas prices decrease.

- **Network Congestion and Transaction Complexity:** Network congestion occurs when the Ethereum network is processing a high volume of transactions. This congestion drives up gas prices, as users compete to prioritize their transactions. Complex transactions, such as those involving smart contracts or dApps, require more computational resources and thus higher gas fees compared to simple ETH transfers.

Additional resources to read:

What is Gas and How is it Used?

What Is a Consensus Mechanism?

▶️ [What are consensus mechanisms?]

A consensus mechanism is the method in which blockchain nodes all come to agreement about whether a piece of data can join the chain. Essentially, they are just a predetermined set of agreements to decide if they should—or shouldn't—add a transaction to their record. This set of agreements allows a blockchain network to reach a conclusion in a decentralized manner, meaning there are no supervisors or admins needed.

What Is Blockchain Consensus For?

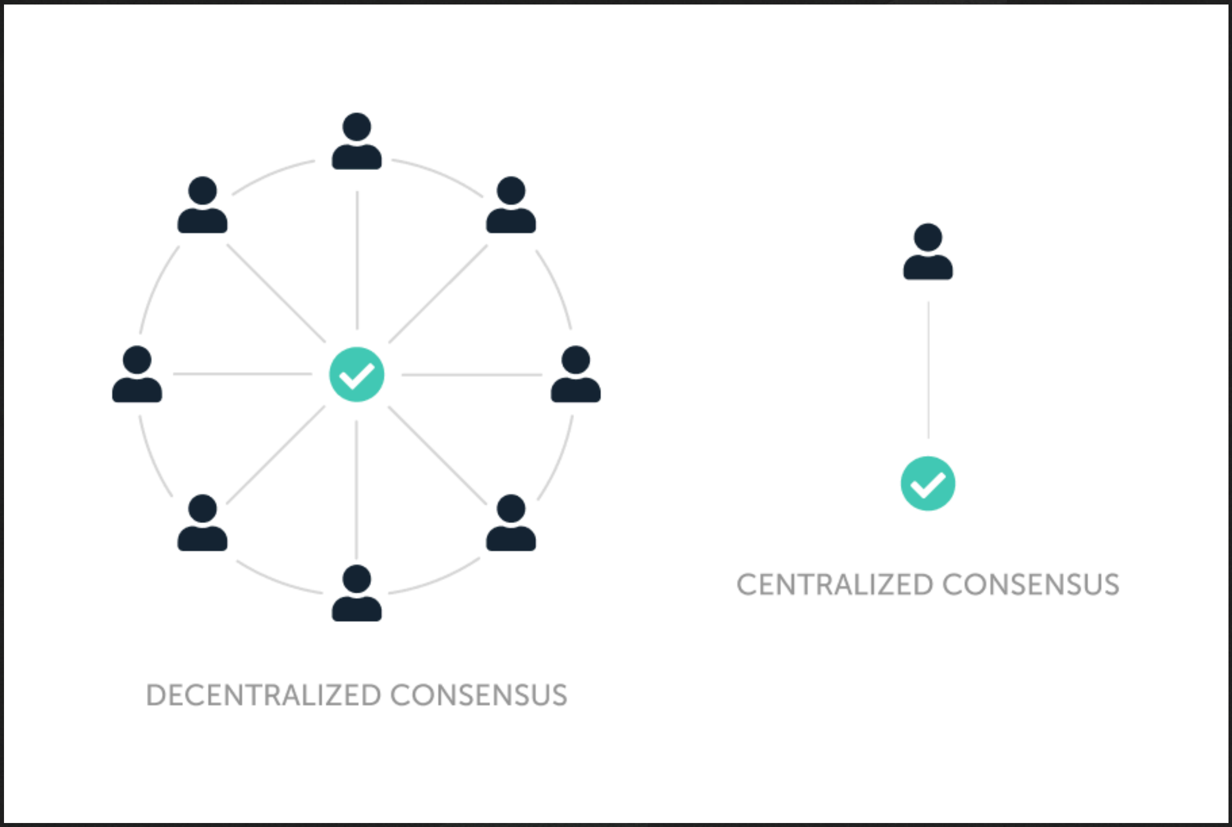
Consensus mechanisms are useful for countless reasons, but the two most notable are; avoiding errors & malicious transactions, and reaching agreement in a decentralized way.

Avoiding Errors & Malicious Transactions

When important data and user balances are on the line, it's extremely important that the blockchain remains free of errors. This way you can guarantee the network only contains genuine transactions. Not only that, using this method will also make sure that malicious transactions are identified. That way, they don't join the chain. Multiple nodes are responsible for adding or rejecting blocks, and this makes it very difficult to sneak a bad transaction through the validation process.

Decentralization

A consensus mechanism is also for decentralization. In short, it allows nodes to come to agreement without a centralized entity's assistance. However, not all blockchain networks using a consensus mechanism are decentralized.



If you've ever heard the term "strength in numbers"—it applies perfectly to blockchains. To explain, the less participants there are in the network to validate and process data, the easier it is for them to reach consensus. As such, a system secured by just a few people is easily corrupted. That means, the more participants there are to reach consensus, the more secure the network is.

How Blockchain Stays Safe

Blockchains are made up of thousands of computers around the world, who are anonymous to each other. These computers all have the same copy of the blockchain. When a new transaction happens, the network checks if it’s valid. Only if more than 50% of the computers agree, the transaction is added.

Because it’s not controlled by one person or company, it’s nearly impossible for anyone to cheat.

A 51% attack is when someone controls more than half of the entire network. For big blockchains like Bitcoin or Ethereum, this kind of attack is almost impossible because:

You would need huge amounts of money to buy or rent enough computers (more than 50% of the ones present in the chain) and a lot of electricity and space to run them .

Moreover, If people see the attack, they can change the network (called a fork) and ignore the attacker. If trust is lost, the coin's value drops, and the attacker loses money too.

To take over a blockchain, it’s more expensive and risky than it's worth. That’s why most attackers don’t even try. It's safer and smarter to be honest in the system.

So now you know what a consensus mechanism is, but what about how it works?

How Do Consensus Mechanisms Work?

Simply, the blockchain only processes a change when a majority of nodes agree that the new piece of data fits with the blockchain's consensus mechanism, that predetermined set of agreements. Since they run using the same framework, every node is expected to reach the same conclusion. And when the majority agree, they reach consensus!

You can think of this as a voting process. If more than half of nodes vote to include it, the transaction will be added to the blockchain. Otherwise, the network discards the transaction.

Then most of the time, a consensus mechanism will involve a way to incentivize honorable participants, and punish bad ones. This is important to keep the network secure, and free from bad actors aiming to exploit the power of processing transactions. Beyond that though, consensus mechanisms can vary greatly from how they are secured to how they process transactions.

The most well known consensus mechanisms of blockchain are Proof of work and Proof of stake, which we will learn in the next task.

12 Consensus Mechanisms + How they Work (Pros/Cons)